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imud — Mathematical Reference (as implemented)

Document purpose. This is an audit reference for the *actual numerical methods executed by the imud source code*, at the release this file ships with. It is deliberately **not** a derivation of the methods imud “should” use, nor a restatement of the papers it draws on: every equation below was transcribed from the code and is annotated with the function, variable, and struct-field names so a reviewer can place each symbol against its implementation. Where the code approximates, deviates from, or specializes its cited source, an **As-implemented** note says so explicitly.

Intended reader. Someone auditing the mathematics itself — an estimation or geophysics specialist — who wants to check the equations and their data flow without reading C. Familiarity with quaternion attitude estimation, Kalman filtering, and spherical-harmonic geomagnetism is assumed.

Scope. All numeric routines in the daemon and its calibration tool: the MEKF attitude/heading filter, accelerometer/magnetometer measurement models, the heave and sea-state estimators, compass-health and engine-vibration metrics, timestamp anchoring, the offline magnetometer / Allan-variance / gyro-temperature calibration fits, and the World Magnetic Model evaluation. The WMM section states the *implementation shape* and cites the defining report rather than re-deriving spherical-harmonic theory.

Source-of-truth files. `src/fusion.c`, `include/fusion.h` (MEKF, heave, sea state); `src/imu.c` (live calibration application, Euler rates, rate of turn, timestamp anchor, engine detector, startup bias); `src/cal_math.c`, `include/cal_math.h` (offline fits, Allan variance); `src/wmm.c`, `include/wmm.h` (geomagnetic model); `src/cal.c` (calibration file I/O).

Citation convention. Each section ends with a **Source** line. Because imud implements standard methods rather than novel mathematics, most sources are the canonical reference for the technique. A citation marked (*code comment*) is the reference named in the source itself; one marked (*canonical*) is the standard literature reference for a technique the code does not cite in-line. See §15 References. Citations the maintainer may wish to verify against the exact edition are flagged in that section.

1. Conventions and reference frames

Frames.

- **Body frame b:** the sensor board axes after the optional mount rotation (§3). NED-compatible: X forward, Y starboard, Z down.

- **Navigation frame NED:** local North-East-Down.

Attitude quaternion. `mekf_t.q = q = [qw, qx, qy, qz]` is a unit Hamilton quaternion, **scalar-first**, representing the **body→NED** rotation:

$$v_{NED} = R(q) v_b, \quad R(q) \in SO(3).$$

The rotation matrix `q_to_R()` (`fusion.c:136`) is

$$R(q) = \begin{bmatrix} 1 - 2(q_y^2 + q_z^2) & 2(q_x q_y - q_w q_z) & 2(q_x q_z + q_w q_y) \\ 2(q_x q_y + q_w q_z) & 1 - 2(q_x^2 + q_z^2) & 2(q_y q_z - q_w q_x) \\ 2(q_x q_z - q_w q_y) & 2(q_y q_z + q_w q_x) & 1 - 2(q_x^2 + q_y^2) \end{bmatrix}.$$

The Hamilton product `q_mul()` (`c = a ⊗ b`, `fusion.c:118`) uses the standard scalar-first convention (Solà eq. 16).

Units. Gyroscope $\text{rad} \cdot \text{s}^{-1}$; accelerometer $\text{m} \cdot \text{s}^{-2}$; magnetometer μT on the wire, converted to **Gauss** ($\times 0.01$) inside the filter; angles rad ; time s . Standard gravity $\mathbf{G}_{MS2} = g = 9.80665 \text{ m s}^{-2}$ (`fusion.c:42`).

Gravity reference. $g_{ref} = [0, 0, 1]$ (unit, NED, Z-down). The accelerometer's *specific force* at rest reads $-g$ on Z; the filter works with the **gravity direction** $z_a = -\hat{a}_b$ (§4.7).

As-implemented. All filter state is single precision (`float`); calibration fits and WMM are double precision. The quaternion is renormalized (`q_normalize`, `fusion.c:126`) after every multiplicative update, guarding unit-norm drift from float round-off.

Source: Solà 2017 (*code comment*); Shuster 1993 for quaternion conventions (*canonical*).

2. Signal flow

One IMU sample traverses, in order:

1. **Driver read** → raw `imu_sample_t{accel[3], gyro[3], temp_c, chip_ts}` in board axes.
2. **Mount rotation** `apply_mount_rot_if_set()` (§3.1) → body axes.
3. **Calibration** `apply_imu_cal()` (§3.2): gyro temperature compensation, accel offset/scale. (Magnetometer: `apply_mag_cal()`, hard/soft iron.)
4. **MEKF** (§4): `mekf_predict` on every IMU sample; `mekf_update_accel` every sample; `mekf_update_mag` per magnetometer sample.
5. **Derived outputs:** Euler angles and heading (§4.10); Euler rates and rate of turn (§5); heave (§6); sea-state statistics (§7); compass-health (§8) and engine (§9) metrics.

Gyro **bias** is *not* removed in step 3; the MEKF estimates and subtracts it (§4.4). Timestamps are reconstructed from the chip counter (§11).

3. Calibration application (live path)

3.1 Mount rotation

`apply_mount_rot_if_set()` (`imu.c:222`) applies a fixed 3×3 board→body matrix $R_{mount} = \text{cfg->mount_rot}$ in place, when `cfg->mount_set`:

$$v \leftarrow R_{mount} v, \quad v \in \{a_b, \omega_b, m_b\}.$$

Applied identically to accel, gyro, and magnetometer vectors. Accumulated in double, stored back as float.

As-implemented. R_{mount} is supplied by configuration (a fixed installation rotation, e.g. yaw 180° for a stern-facing board); `imu` does not estimate it. No orthonormalization is performed on the configured matrix.

3.2 Inertial calibration

`apply_imu_cal()` (`imu.c:141`), per axis i :

- **Gyro temperature compensation** (when `cal->has_gyro_temp`):

$$\omega_i \leftarrow \omega_i - c_i (T - T_{ref}),$$

with $c_i = \text{gyro_temp_coeff}[i]$, $T = \text{temp_c}$, $T_{ref} = \text{gyro_temp_ref_c}$ (fit in §12.6).

- **Accel offset/scale** (when `cal->has_accel`):

$$a_i \leftarrow (a_i - o_i) s_i,$$

with $o_i = \text{accel_offset}[i]$, $s_i = \text{accel_scale}[i]$.

3.3 Magnetometer hard/soft-iron

`apply_mag_cal()` (`imu.c:159`), when `cal->has_mag`:

$$m_{cal} = S_{soft} (m_{raw} - b_{hard}),$$

with $b_{hard} = \text{mag_hard_iron}[3]$, $S_{soft} = \text{mag_soft_iron}[3][3]$. The correction parameters are produced offline (§12.2–12.3).

As-implemented. The accel model is diagonal scale + offset (no off-diagonal cross-axis / misalignment terms). The magnetometer model is a full 3×3 soft-iron matrix times a hard-iron offset, but that matrix is fitted from a **2-D** swing (§12.3), so its out-of-plane (dip) accuracy is limited — which is exactly why the filter defaults to heading-only magnetometer fusion (§4.8).

Source: standard sensor error model; Titterton & Weston 2004 §12 (*canonical*).

4. Multiplicative Extended Kalman Filter

The attitude/heading core is an **error-state (multiplicative) EKF**. The nominal state carries the full quaternion and gyro bias; the Kalman filter operates on a minimal 6-D **error state**, and corrections are injected multiplicatively so the quaternion never leaves the unit sphere.

4.1 State and covariance

Nominal state (`mekf_t`, `fusion.h:31`):

- q — unit quaternion, body→NED (`q[4]`).
- b — gyro bias, $\text{rad} \cdot \text{s}^{-1}$ (`bias[3]`).

Error state (never stored explicitly; the axis of P):

$$\delta x = [\delta\theta \ (3) \mid \delta b \ (3)] \in \mathbb{R}^6,$$

$\delta\theta$ = small-angle rotation error (rad), δb = bias error. Covariance $P \in \mathbb{R}^{6 \times 6}$ (`P[6][6]`), symmetric PD; top-left 3×3 is attitude-error covariance (rad^2), bottom-right 3×3 is bias-error covariance ($(\text{rad} \cdot \text{s}^{-1})^2$).

The nominal/error relationship is the **right** (local) perturbation $q_{true} = \hat{q} \otimes \delta q(\delta\theta)$, established by the sign of the measurement Jacobian in §4.5.

Source: Solà 2017 §5.4 (*code comment*); Markley & Crassidis 2014 §6.1; Trawny & Roumeliotis 2005 (*canonical*).

4.2 Initialization — `mekf_init()` (`fusion.c:400`)

Nominal: $q = [1, 0, 0, 0]$, $b = \text{gyro_bias_init}$ (from the startup still window, §10; may be 0).

Initial covariance (diagonal):

$$P_0[0:3] = (0.175)^2 \text{ rad}^2 (\approx 10^\circ), \quad P_0[3:6] = (0.001)^2 (\text{rad s}^{-1})^2.$$

Discrete process-noise variances, step $dt = 1/\text{ODR}$:

$$Q_g = N_g^2 dt, \quad Q_b = N_b^2 dt,$$

with $N_g = \text{mekf_gyro_noise}$, $N_b = \text{mekf_gyro_bias}$ (datasheet noise densities). Stored as $\mathbf{f} \rightarrow \mathbf{Qg}$, $\mathbf{f} \rightarrow \mathbf{Qb}$.

Measurement-noise variances:

$$R_a = \left(\frac{N_a}{g} \right)^2 \text{ ODR}, \quad R_m = N_m^2 f_{s,\text{mag}},$$

$\mathbf{f} \rightarrow \mathbf{Ra}$ (`fusion.c:448`) in normalized (gravity-direction) units, $\mathbf{f} \rightarrow \mathbf{Rm}$ (:449) in Gauss², with $N_a = \text{mekf_accel_noise}$, $N_m = \text{mekf_mag_noise}$, $f_{s,\text{mag}} = \text{mag_odr_hz}$.

Derived thresholds: accel skip band $[1 - s_k, 1 + s_k]$ with $s_k = \text{accel_skip_thresh}$; $\text{mag_reject_sq} = (\text{mag_reject_gauss})^2$; convergence threshold $\text{conv_thresh} = 3 (0.5^\circ \text{ in rad})^2$ (:471); m_ref EMA gain $mref_alpha = 1/(\tau_{mref} f_{s,\text{mag}})$ with $\tau_{mref} = 300 \text{ s}$ (:467).

As-implemented. Q_g, Q_b are *per nominal step*; during prediction they are rescaled by the actual dt (§4.4). The bias process is modeled as a random walk whose density N_b is a **tuning constant deliberately held above** the measured in-run bias instability (see `docs/capture.md`); it is not driven by the Allan-variance characterization of §12.5.

Source: Solà 2017 §4.1 for discrete process noise (*code comment*).

4.3 Alignment — `mekf_align()` (`fusion.c:494`)

Deterministic initial attitude from one static accel+mag pair (a tilt-then-heading decomposition, TRIAD-family).

Tilt from accelerometer. With $\hat{g}_b = -a_b/\|a_b\|$ the gravity direction in body (guard: reject if $\|a_b\| < 0.5g$):

$$\phi = \text{atan2}(\hat{g}_{b,y}, \hat{g}_{b,z}), \quad \theta = \text{atan2}(-\hat{g}_{b,x}, \sqrt{\hat{g}_{b,y}^2 + \hat{g}_{b,z}^2}).$$

Tilt quaternion $q_{\text{tilt}} = q_\theta \otimes q_\phi$ built directly from half-angles (`fusion.c:508`).

Heading from magnetometer. Rotate m_b by $R(q_{\text{tilt}})$ into the tilt-levelled frame; with horizontal components m_x, m_y there,

$$\psi = \text{atan2}(-m_y, m_x).$$

Full attitude $q = q_\psi \otimes q_{\text{tilt}}$, $q_\psi = [\cos \frac{\psi}{2}, 0, 0, \sin \frac{\psi}{2}]$.

Magnetic reference. On first alignment, $\mathbf{m_ref}$ (NED, Gauss) is set from the measurement rotated to NED and scaled $\mu\text{T} \rightarrow \text{Gauss}$:

$$m_{ref} = 0.01 R(q) m_b.$$

As-implemented / audit note. The heading sign is $\text{atan2}(-m_y, m_x)$, **not** $\text{atan2}(+m_y, m_x)$; the levelled field is the true NED field rotated by $-\psi$, so the negative sign recovers vessel yaw. (The $+\mathbf{m}_y$ form returns $-\psi$, mirrored about north — a historical bug invisible when aligning while pointing north; see the in-code comment at `fusion.c:519`.) The initial `m_ref` inherits any alignment tilt error in its magnitude/dip, which the quiescence-gated EMA (§4.8) and the WMM invariants (§4.9) later remove; its horizontal **direction** is the heading datum and is never subsequently adapted.

Source: tilt/heading coarse alignment — Farrell 2008 §10; TRIAD lineage Shuster & Oh 1981 (*canonical*).

4.4 Prediction — `mekf_predict()` (`fusion.c:551`)

Per IMU sample, with measured interval dt (from hardware timestamps, §11; falls back to nominal $1/\text{ODR}$).

Bias-corrected rate: $\omega = s.\text{gyro} - b$.

Quaternion propagation by the exponential map (`q_from_rotvec`, `fusion.c:149`):

$$q \leftarrow q \otimes \exp\left(\frac{1}{2} [0, \omega dt]\right), \quad \exp(\phi) = \begin{bmatrix} \cos \frac{|\vartheta|}{2}, & \frac{\sin(|\vartheta|/2)}{|\vartheta|} \vartheta \end{bmatrix},$$

$\vartheta = \omega dt$, with the small-angle branch $\delta q \approx [1, \frac{1}{2}\vartheta]$ for $|\vartheta| < 10^{-7}$. Renormalized after.

Covariance propagation $P \leftarrow \Phi P \Phi^\top + Q_d$, first-order discrete transition (`fusion.c:588`):

$$\Phi = \begin{bmatrix} I_3 - [\omega]_\times dt & -I_3 dt \\ 0_3 & I_3 \end{bmatrix}, \quad Q_d = \text{diag}(Q_g \frac{dt}{dt_0} I_3, Q_b \frac{dt}{dt_0} I_3),$$

$[\omega]_\times$ the skew-symmetric cross-product matrix; the noise is rescaled by dt/dt_0 (dt_0 = nominal step) so variance grows linearly with the real interval. P is symmetrized after (:610). Convergence flag $\text{tr}(P[0:3]) < \text{conv_thresh}$ (:619).

As-implemented. Prediction uses the plain $\Phi P \Phi^\top + Q$ form (not Joseph — the Joseph stabilization is applied on the *updates*, §4.5). Φ is a first-order (zero-hold) approximation of $\exp(F_c dt)$; valid because $\|\omega\|dt \ll 1$ at supported ODRs. The $[\omega]_\times$ sign in $\Phi_{[0:3,0:3]}$ is that of the right-perturbation error convention.

Source: Solà 2017 eq. 259 (integration), eq. 268 (covariance) (*code comment*).

4.5 Generic vector measurement update — `eskf_update()` (`fusion.c:185`)

Measurement of a known NED reference observed in body: predicted h , actual z , isotropic noise R_{noise} , gate `chi2_gate`.

Jacobian (3×6), attitude block only:

$$H = [[h]_\times \mid 0_3], \quad [h]_\times = \begin{bmatrix} 0 & -h_2 & h_1 \\ h_2 & 0 & -h_0 \\ -h_1 & h_0 & 0 \end{bmatrix}.$$

Innovation covariance and gain:

$$S = H P H^\top + R_{\text{noise}} I_3, \quad K = P H^\top S^{-1},$$

S^{-1} by Cramer's rule (`m33_inv`, `fusion.c:73`; singular $\Rightarrow$ skip).

Innovation $\nu = z - h$, with **robust (Huber-style) gating** on the Mahalanobis distance $d^2 = \nu^\top S^{-1} \nu$:

$$\begin{cases} \text{reject the update} & d^2 > 9\gamma \\ \nu \leftarrow \nu \sqrt{\gamma/d^2} & \gamma < d^2 \leq 9\gamma \\ \nu \text{ unchanged} & d^2 \leq \gamma \end{cases}$$

$\gamma = \text{chi2_gate}$. The middle branch caps the innovation's influence at the gate distance rather than discarding it.

Correction and injection:

$$\delta x = K \nu, \quad q \leftarrow q \otimes \exp(\delta \theta), \quad b \leftarrow b + \delta b,$$

$\delta \theta = \delta x[0:3]$, $\delta b = \delta x[3:6]$; quaternion renormalized.

Covariance — Joseph form (`fusion.c:284`):

$$P \leftarrow (I - KH) P (I - KH)^\top + R_{noise} K K^\top,$$

then symmetrized. (Isotropic $R \Rightarrow K R K^\top = R_{noise} K K^\top$.)

As-implemented / audit note. The error is *reset* into the quaternion each update but P is **not** rotated by the reset Jacobian $G = I - \frac{1}{2}[\delta \theta]_\times$ (the common first-order reset). At the per-sample correction magnitudes here $G \approx I$; the omission is a deliberate first-order simplification. The Huber cap replaces a hard χ^2 reject so that wave-orbital acceleration (which swings gravity *direction* while keeping $|a| \approx g$) deweights rather than starves the filter; only gross outliers ($d^2 > 9\gamma$) are rejected. Because S contains P , the cap is naturally inactive during acquisition (large P) and engages only once confident — no explicit convergence gate needed.

Source: EKF update Kalman 1960; Joseph form Bucy & Joseph 1968; robust innovation capping Huber 1964; Mahalanobis gating Bar-Shalom et al. 2001 (*canonical*). Jacobian sign Solà 2017 §7 (*code comment*).

4.6 Scalar heading update — `eskf_update_yaw()` (`fusion.c:337`)

Heading-only correction (a 1-D measurement). Innovation y (rad, wrapped to $\pm\pi$), variance R_{noise} . The Jacobian projects the error onto the NED-down axis expressed in body:

$$H = \begin{bmatrix} -R[2][:] & 0_3 \end{bmatrix} \in \mathbb{R}^{1 \times 6},$$

$R[2][:]$ = third row of $R(q)$. Then

$$S = H P H^\top + R_{noise}, \quad K = P H^\top / S,$$

with the same Huber policy on $d^2 = y^2 / S$ against $\gamma_\psi = 6.63$ (χ_1^2 99%), rank-1 Joseph covariance update.

Source: scalar-measurement EKF specialization of §4.5 (*canonical*).

4.7 Accelerometer update — `mekf_update_accel()` (`fusion.c:628`)

Speed-aided centripetal correction. In a turn the accelerometer senses $a = a_{platform} - g$, and with speed-over-ground v (body $\approx [v, 0, 0]$, leeway neglected) the centripetal term is $\omega \times v = [0, \omega_z v, -\omega_y v]$. When `speed_mps` > 0.1:

$$a_y \leftarrow a_y - \omega_z v, \quad a_z \leftarrow a_z + \omega_y v,$$

$\omega_{y,z}$ bias-corrected. `speed_mps=0` (no GPS) $\Rightarrow$ no-op.

Quiescence EMA (updated every sample, incl. gated), $\tau \approx 2$ s:

$$q_{quiet} \leftarrow q_{quiet} + ((a_g - 1)^2 - q_{quiet}) \frac{dt}{2}, \quad a_g = \|a\|/g.$$

Skip gate. If $a_g \notin [\text{accel_skip_lo}, \text{accel_skip_hi}]$, skip (linear acceleration).

Measurement. Gravity direction $z = -a/\|a\|$; prediction $h = R^\top g_{ref} = R[2][:]$ (third row). Effective noise $R_{eff} = R_a \cdot \text{Ra_scale}$. Then `eskf_update(h, z, R_eff, 11.34)` with $\gamma = 11.34$ (χ_3^2 99%). The gravity direction z is stashed as `g_body` (attitude-independent dip reference for §4.8).

As-implemented. `Ra_scale` is set to 4 by the fusion thread while engine vibration is detected (§9) — vibration is high-frequency, near-zero-mean, so deweighting (not gating) is correct. Wave *direction* disturbance is

handled by the Huber cap (§4.5), **not** by magnitude-based deweighting, which benchmarking showed rectifies wave-phase-correlated error into an attitude/bias offset. The centripetal model assumes zero leeway and no vertical velocity.

Source: specific-force / coordinated-turn model Titterton & Weston 2004; Farrell 2008 (*canonical*).

4.8 Magnetometer update — `mekf_update_mag()` (`fusion.c:697`)

Measurement in Gauss ($m = 0.01 m_{\mu T}$). Predicted body field $h_{raw} = R^\top m_{ref}$, magnitude $|h_{raw}|$.

Normalization. Both predicted and measured vectors are unit-normalized before the update; the mag noise is rescaled $R_{m,n} = R_m / |h_{raw}|^2$ to preserve the optimal gain. (Rationale: in Gauss² units $\det S \approx R_m^3 \approx 4 \times 10^{-15}$ falls below the inverse's singularity threshold once P_{att} is small; normalizing makes S dimensionless and well-conditioned.) Residual $\nu = z - h$, $\text{res_sq} = \|\nu\|^2$.

Compass-health metrics — computed **before** the gates (§8).

Gates. Magnitude ratio $r = |m| / |h_{raw}|$ must lie in $[0.5, 2]$; direction $\text{res_sq} < 4$ ($\leq 90^\circ$ correction); and, once converged, the tight anomaly gate $\text{res_sq} > \text{mag_reject_sq} / |h_{raw}|^2 \Rightarrow \text{skip}$.

Reference adaptation (m_ref EMA), magnitude + dip only. When `mref_alpha` > 0, `g_body_valid`, $\text{res_sq} < 0.1$, and $q_{quiet} < 2 \times 10^{-4}$ (platform calm), adapt the field magnitude and dip from **attitude-independent invariants**: $\sin(\text{dip}) = (m \cdot g_{body}) / |m|$,

$$m_{ref}^h \leftarrow m_{ref}^h (1 + \alpha_{mref} (|m| \cos \text{dip} / m_{ref}^h - 1)), \quad m_{ref,z} \leftarrow m_{ref,z} + \alpha_{mref} (|m| \sin \text{dip} - m_{ref,z}).$$

The horizontal **direction** is never touched.

Update. If `mag_yaw_only` (marine default): rotate m to NED, form the heading innovation

$$y = \text{atan2}(m_y^{NED}, m_x^{NED}) - \text{atan2}(m_{ref,y}, m_{ref,x}) \text{ (wrapped)},$$

noise $R_\psi = R_m / (m_{ref}^h)^2$, and call the scalar update §4.6. Otherwise call the full vector update §4.5 with $(h, z, R_{m,n}, 11.34)$. A near-vertical field ($m^h < 0.2 |h_{raw}|$) carries no heading information and is skipped.

As-implemented / audit note. Adapting **only** magnitude and dip is deliberate: the horizontal direction *is* the heading reference, so learning it from the filter's own attitude would feed the heading estimate back into its own datum and let both random-walk together (gauge drift). The adaptation targets are body-frame invariants (total magnitude; dip from the mag-gravity angle), so the filter's attitude never enters. Heading-only fusion is the default because the soft-iron matrix is fitted from a 2-D swing (§3.3, §12.3) and its dip channel is least trustworthy.

Source: heading-only / tilt-compensated magnetometer fusion — standard marine AHRS practice, Farrell 2008 (*canonical*); gauge-feedback avoidance is imud-specific design (see in-code rationale `fusion.c:781`).

4.9 WMM reference invariants — `mekf_set_mref_invariants()` (`fusion.c:474`)

Given WMM horizontal magnitude H and vertical Z (Gauss) at a known position (from §13), rescale `m_ref` to those invariants while preserving horizontal direction: with $m_h = \sqrt{m_{ref,x}^2 + m_{ref,y}^2}$ and $s = H / m_h$,

$$m_{ref,x} \leftarrow s m_{ref,x}, \quad m_{ref,y} \leftarrow s m_{ref,y}, \quad m_{ref,z} \leftarrow Z.$$

No-op before alignment or if $H \leq 0$.

Source: imud-specific (WMM supplies the invariants of §13).

4.10 State extraction — `mekf_get_state()` (`fusion.c:1030`)

Euler angles from $R(q)$ (NED 3-2-1 aerospace):

$$\theta = \arcsin(-R[2][0]), \quad \phi = \operatorname{atan2}(R[2][1], R[2][2]), \quad \psi = \operatorname{atan2}(R[1][0], R[0][0]).$$

Magnetic heading ψ wrapped to $[0, 360^\circ)$. Attitude covariance (`cov[9]`) = $P[0:3, 0:3]$; bias variance = $\operatorname{diag} P[3:6]$; `quiescence` = q_{quiet} . `rate_of_turn` is left 0 here and filled by the fusion thread (§5).

Source: quaternion→Euler, Diebel 2006 (*canonical*).

4.11 Reconfigure — `mekf_reconfigure()` (`fusion.c:1005`)

Recomputes Q_g, Q_b, R_a, R_m , the skip band, `mag_reject_sq`, and `mag_yaw_only` from a new config on hot-reload; q, P, b, dt untouched (the filter keeps running).

5. Euler rates and rate of turn

Computed in the fusion thread (`imu.c:1004–1051`) from the bias-corrected body rate $\omega = s.\text{gyro} - b$ and the current Euler angles, using the inverse of the 3-2-1 kinematic relation.

Roll rate (for sea-state, §7), with a near-gimbal-lock fallback:

$$\dot{\phi} = \omega_x + \tan \theta (\omega_y \sin \phi + \omega_z \cos \phi), \quad \dot{\phi} = \omega_x \text{ if } |\cos \theta| \leq 0.2.$$

Pitch rate (no singularity):

$$\dot{\theta} = \omega_y \cos \phi - \omega_z \sin \phi.$$

Rate of turn (heading rate, output in $\text{deg} \cdot \text{min}^{-1}$):

$$\dot{\psi} = \frac{\omega_y \sin \phi + \omega_z \cos \phi}{\cos \theta}, \quad \dot{\psi} = \omega_z \text{ if } |\cos \theta| \leq 0.2, \quad \text{ROT} = \dot{\psi} \cdot \frac{180}{\pi} \cdot 60.$$

As-implemented. ROT is the *Euler yaw rate*, not raw body-Z gyro: at 20° roll the body-Z rate under-reads heading rate and couples in pitch rate, which autopilots notice. Near $\pm 90^\circ$ pitch ($|\cos \theta| \leq 0.2$) the code falls back to the body-axis rate to avoid division blow-up.

Source: 3-2-1 Euler kinematics, Titterton & Weston 2004 §3.6; Diebel 2006 (*canonical*).

6. Heave estimator — `heave_update()` (`fusion.c:880`)

Vertical displacement from vertical acceleration, by leaky double integration followed by a true first-order high-pass.

Vertical acceleration (NED down, zero at rest):

$$a_D = (R(q) a_b)_D + g = R[2][:] \cdot a_b + g.$$

Leaky double integration, leak $\ell = dt/\tau$:

$$v \leftarrow (v + a_D dt)(1 - \ell), \quad d \leftarrow (d + v dt)(1 - \ell).$$

Output high-pass (exact zero at DC), $\alpha_{hp} = \tau/(\tau + dt)$:

$$y \leftarrow \alpha_{hp} (y + d - d_{\text{prev}}), \quad d_{\text{prev}} \leftarrow d,$$

heave (positive **up**) = $-y$. **heave_rate** output = $-v$. Settled after $\geq 10\tau$ (**settled** flag).

As-implemented / audit note. Two leaky integrators bound drift but retain a DC gain of $\sim \tau^2$ (≈ 144 m per $\text{m} \cdot \text{s}^{-2}$ at the default $\tau = 12$ s); any slow residual (accel bias, small tilt error, start-up transient) would appear at metre scale. The output high-pass has a **structural** zero at DC, killing those residuals rather than estimating them. Passband amplitude error $\approx 2\%$ at 8 s, $\approx 4\%$ at 12 s for $\tau = 12$ s (band-limited: very long-period swell is attenuated). $\tau = 0$ disables. The estimate is a filtered kinematic quantity, not a statistically-optimal one.

Source: leaky integrator + first-order high-pass are standard discrete-time filter forms, Oppenheim & Schaffer 2010; the double-integration- plus-high-pass heave technique is long-standing in wave-buoy practice (Datawell / Longuet-Higgins tradition) (*canonical*).

7. Sea-state statistics — **seastate_*** (**fusion.c:907–1003**)

Windowed spectral moments over the heave, roll, and pitch oscillations via exponentially-weighted mean/variance pairs — no FFT, no sample storage.

EW mean/variance recursion (**ew_stat**, **fusion.c:937**), $\alpha = dt/\tau$:

$$\mu \leftarrow \mu + \alpha d, \quad \sigma^2 \leftarrow \sigma^2 + \alpha((1 - \alpha)d^2 - \sigma^2), \quad d = x - \mu.$$

Applied to six signals: heave, heave-rate, roll, roll-rate, pitch, pitch-rate. Fed **only while heave is settled** (**seastate_update**, **fusion.c:944**).

Outputs (from the variances; $m_0 = \text{var}(x)$, $m_2 = \text{var}(\dot{x})$):

$$\begin{aligned} H_s &= 4\sqrt{m_0^{heave}}, & T_z &= 2\pi\sqrt{\frac{m_0^{heave}}{m_2^{heave}}}, & T_{roll} &= 2\pi\sqrt{\frac{m_0^{roll}}{m_2^{roll}}}, \\ A_{roll} &= 2\sqrt{m_0^{roll}}, & A_{pitch} &= 2\sqrt{m_0^{pitch}}, & T_{pitch} &= 2\pi\sqrt{\frac{m_0^{pitch}}{m_2^{pitch}}}. \end{aligned}$$

Guards: all return 0 until settled ($\geq 2\tau$); periods additionally return 0 below the oscillation floors $\sigma(\text{heave}) < 2$ cm (**SEASTATE_MIN_HEAVE_SIG**) or $\sigma(\text{angle}) < 0.3^\circ$ (**SEASTATE_MIN_ANGLE_SIG**), or non-positive rate variance.

As-implemented / audit note. $H_s = 4\sqrt{m_0}$ is the significant wave height for a narrow-band Gaussian sea (Longuet-Higgins). The period identity $T = 2\pi\sqrt{m_0/m_2}$ is Rice's mean up-crossing period from the zeroth and second spectral moments — **exact for a pure sine** ($\text{var} = A^2/2$, $\text{var}(\dot{x}) = A^2\omega^2/2$, $\text{ratio} = 1/\omega^2$) and an estimator for a real seaway. Amplitudes are reported as **significant single amplitudes** (2σ , seakeeping convention); wave height is the significant **double** amplitude (4σ). The EW variance is biased toward the window's recent history and toward the EW mean (so a steady heel / trim is removed from roll/pitch, appearing in μ , not σ). The default $\tau = 120$ s is far shorter than the 10–20 min oceanographic record; it favours a responsive live display. Roll/pitch **rates** are Euler rates (§5), not body rates.

Source: $H_s = 4\sqrt{m_0}$ Longuet-Higgins 1952; spectral-moment up-crossing period Rice 1944–45; engineering treatment Tucker & Pitt 2001; roll-period/seakeeping convention Lloyd 1989 (*canonical*). EW variance recursion, West 1979 / Finch 2009 (*canonical*).

8. Compass-health diagnostics — **mekf_update_mag()** (**fusion.c:749**)

Two EMAs ($\alpha = 1/3000$, $\tau \approx 30$ s at 100 Hz mag ODR), updated **before** the rejection gates so that gated-out anomalies still register:

Field-magnitude anomaly (attitude-independent): $\text{mag_anom} \leftarrow \text{mag_anom} + \alpha \left(\frac{|\text{h_raw}|}{|\text{h_ref}|}, |\text{h_raw}| \right)$

- mag_anom (right). mag_anom

Heading residual (mode-independent; $m^{NED} = R(q) m$):

$$y = \text{atan2}(m_y^{NED}, m_x^{NED}) - \text{atan2}(m_{ref,y}, m_{ref,x}) \text{ (wrapped to } \pm\pi), \quad \text{mag_resid} \leftarrow \text{mag_resid} + \alpha(|y| - \text{mag_resid}),$$

gated on both horizontal magnitudes exceeding $0.2|h_{raw}|$.

As-implemented. Diagnostics only — they never modify the filter. Fed from pre-gate innovations because the rejected samples are precisely the anomalies these metrics exist to surface. The 30 s time constant scales inversely with a non-standard mag rate (acceptable for a health indicator).

Source: exponential moving average, standard (*canonical*).

9. Engine-vibration detector — `ism_reader_thread()` (`imu.c`)

EMA of squared specific-force deviation from 1 g, $\alpha = 0.01$ ($\tau \approx 1$ s at 100 Hz burst rate), per accepted sample:

$$e \leftarrow e + \alpha((\|a\| - g)^2 - e), \quad \text{engine_on} = [e > \text{engine_vibration_g2}].$$

`engine_on` raises the accel noise $\times 4$ and widens the skip band (§4.7).

Source: EMA threshold detector, standard (*canonical*).

10. Startup gyro-bias estimation — `fusion_thread()` (`imu.c:729`)

Mean of the gyro over a still window ($N = \text{gyro_bias_sec} \cdot \text{ODR}$ samples):

$$\hat{b}_k = \frac{1}{N} \sum_{i=1}^N \omega_{i,k}.$$

A per-axis motion check computes the sample standard deviation $\sigma_k = \sqrt{\omega^2 - \bar{\omega}^2}$; if $\max_k \sigma_k > 0.00873$ rad $\cdot$ s $^{-1}$ ($0.5^\circ \cdot$ s $^{-1}$) the window is **doubled once** (a longer average spans more wave cycles). The result seeds `mekf_init` (§4.2); the MEKF refines it online regardless.

Source: sample mean/variance, elementary (*canonical*).

11. Timestamp anchoring and per-sample dt — `imu.c:173–207`

A single anchor (`chip_ticks`, `wall_ns`, `tai_ns`, `gen`) maps the sensor's free-running counter to system time. Per sample (`chip_to_wall`, `imu.c:197`), with `tick_ns` the counter period:

$$\text{offset} = (\text{chip_ts} - \text{chip_ticks}) \cdot \text{tick_ns}, \quad t_{\text{wall}} = \text{wall_ns} + \text{offset},$$

t_{tai} likewise. Subtraction is unsigned 32-bit (wrap-safe within one anchor interval). The anchor is refreshed on first read, every 60 s with a hardware counter, or every burst without one.

Per-sample interval for prediction (`imu.c`): the wall-time delta between consecutive samples of the same anchor generation, **clamped to** $[0.5, 2] \times \text{nominal}$; outside that (FIFO gap, anchor reset) the nominal $1/\text{ODR}$ is used. This keeps oscillator tolerance (few % on the chip clock) from scaling integrated rotation while rejecting gap artefacts.

As-implemented. Linear (constant-rate) interpolation between anchors; no clock-drift model beyond the 60 s re-anchor. `tick_ns=0` for chips without a hardware timestamp degenerates the offset to 0 (the anchor wall-time is then the per-sample time).

Source: imud-specific timestamp reconstruction.

12. Offline calibration fits (`imud-cal`, `cal_math.c`)

12.1 Linear solver — `gauss4()` (`cal_math.c:24`)

Gaussian elimination with **partial pivoting** on the augmented 4×5 system, then back-substitution. Returns -1 if any pivot $< 10^{-12}$. Double precision.

Source: Golub & Van Loan 2013 §3.4 (*canonical*).

12.2 Hard-iron sphere fit — `sphere_fit()` (`cal_math.c:125`)

Algebraic (linearized) sphere fit. The sphere $(x - c_x)^2 + (y - c_y)^2 + (z - c_z)^2 = r^2$ is linearized as

$$x^2 + y^2 + z^2 = 2c_x x + 2c_y y + 2c_z z + (r^2 - \|c\|^2),$$

and the normal equations $A^\top A p = A^\top b$ are accumulated incrementally (`sphere_add`, `cal_math.c:58`) over sums $\Sigma x, \Sigma x^2, \Sigma xy, \Sigma xr^2, \dots$. Solved by `gauss4`:

$$A^\top A = \begin{bmatrix} 4S_{xx} & 4S_{xy} & 4S_{xz} & 2S_x \\ 4S_{xy} & 4S_{yy} & 4S_{yz} & 2S_y \\ 4S_{xz} & 4S_{yz} & 4S_{zz} & 2S_z \\ 2S_x & 2S_y & 2S_z & n \end{bmatrix}, \quad A^\top b = \begin{bmatrix} 2S_{xr} \\ 2S_{yr} \\ 2S_{zr} \\ S_{xx} + S_{yy} + S_{zz} \end{bmatrix},$$

center $c = p[0:3]$, $r = \sqrt{p_3 + \|c\|^2}$. Center = hard-iron offset; r = reference field magnitude for §12.3.

As-implemented. This is the *algebraic* fit (minimizes an algebraic residual, not orthogonal geometric distance); fast, closed-form, and adequate for well-sampled swings, but statistically biased for sparse/noisy data versus a geometric fit.

Source: algebraic sphere/magnetometer fit, Renaudin et al. 2010; Coope 1993 (*canonical*).

12.3 Soft-iron 2-D ellipse fit — `ellipse_fit()` (`cal_math.c:78`)

Conic least-squares on the horizontal magnetometer locus $Ax^2 + Bxy + Cy^2 = 1$ (regressors x^2, xy, y^2 ; normal equations from $S_{40}, S_{31}, S_{22}, S_{13}, S_{04}, S_{20}, S_{11}, S_{02}$ via `ellipse_add`, solved by `gauss4` with an identity pad row). Requires the conic $M = \begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix}$ positive-definite ($A > 0, \det > 0$).

Soft-iron matrix $S = r \cdot M^{1/2}$ via the 2×2 symmetric eigendecomposition: eigenvalues $\lambda_{1,2} = \frac{A+C}{2} \pm \sqrt{\frac{1}{4}(A-C)^2 + \frac{1}{4}B^2}$, unit eigenvector v for λ_1 , and

$$S = s_1 v v^\top + s_2 v_\perp v_\perp^\top, \quad s_{1,2} = r \sqrt{\lambda_{1,2}}, \quad v_\perp = (-v_y, v_x).$$

As-implemented / audit note. The soft-iron correction is **2-D** (X–Y plane only). This is a deliberate specialization for the heading-only fusion default (§4.8): the swing is a horizontal circle, so the vertical/dip channel is not calibrated. `ellipse_fit` includes a diagonal fallback when the conic is degenerate. The full `mag_soft_iron[3][3]` applied live (§3.3) therefore has its Z couplings from configuration/identity, not from this fit.

Source: algebraic conic/ellipse fit, Fitzgibbon, Pilu & Fisher 1999; magnetometer soft-iron, Renaudin et al. 2010 (*canonical*).

12.4 Heading-circle coverage — `cal_cov_mark()` (`cal_math.c:151`)

Guided-swing coverage: the heading circle is split into `nsec` sectors; the sample (x, y) relative to center (c_x, c_y) is binned by

$$s = \left\lfloor \frac{\text{atan2}(y - c_y, x - c_x) \bmod 2\pi}{2\pi/\text{nsec}} \right\rfloor.$$

`cal_cov_count` sums the marked sectors. Pure bookkeeping (no fit).

12.5 Allan variance — `allan_deviation()` (`cal_math.c:170`)

Overlapping Allan deviation. With the cumulative integral $\theta[k] = \sum_{j < k} x_j dt$ and octave cluster lengths $m = 1, 2, 4, \dots$ ($\tau = m dt$):

$$\sigma^2(\tau) = \frac{1}{2\tau^2 (N - 2m + 1)} \sum_{k=0}^{N-2m} (\theta[k+2m] - 2\theta[k+m] + \theta[k])^2.$$

Characterization (`allan_characterize`, `cal_math.c:203`): white-noise density from the shortest cluster,

$$N = \sigma(\tau_0) \sqrt{\tau_0},$$

bias instability from the curve minimum,

$$B = 0.664 \cdot \min_{\tau} \sigma(\tau).$$

As-implemented. Octave (not fully-overlapping all- τ) spacing; N is read at τ_0 (assumes white noise dominates there); B uses the IEEE-952 factor 0.664 at the deviation minimum (an upper bound if the flat region is not reached). These characterize the sensor; they do **not** feed the filter tuning (§4.2 note).

Source: Allan 1966; overlapping estimator Riley 2008 (NIST SP 1065); N/B extraction and 0.664, IEEE Std 952 (*code comment: "IEEE 952"*).

12.6 Gyro-bias / temperature fit — `gyro_temp_fit()` (`cal_math.c:227`)

Ordinary least-squares line of gyro bias vs. temperature about a reference T_{ref} : with $t_i = T_i - T_{ref}$,

$$\text{coeff} = \frac{n \sum t_i y_i - \sum t_i \sum y_i}{n \sum t_i^2 - (\sum t_i)^2}, \quad \text{bias_ref} = \frac{\sum y_i - \text{coeff} \sum t_i}{n}.$$

Rejects fits with temperature span < 1 °C. `coeff` → `gyro_temp_coeff`, applied live in §3.2.

Source: ordinary least squares, elementary (*canonical*).

13. World Magnetic Model — `wmm_field_ned()` (`wmm.c:119`)

Degree/order 12 spherical-harmonic geomagnetic field. This section states the implemented computation; the defining theory is the WMM report cited below.

Secular variation to the decimal year t (`wmm_decimal_year`, `wmm.c:73`), $\Delta t = t - t_{epoch}$:

$$g_n^m(t) = g_n^m + \dot{g}_n^m \Delta t, \quad h_n^m(t) = h_n^m + \dot{h}_n^m \Delta t.$$

Geodetic (WGS-84) → geocentric spherical. With prime-vertical radius $N = a / \sqrt{1 - e^2 \sin^2 \varphi}$ ($a = 6378137$ m, e^2 from $f = 1/298.257223563$):

$$p_x = (N + \text{alt}) \cos \varphi, \quad p_z = (N(1 - e^2) + \text{alt}) \sin \varphi,$$

$$r = \sqrt{p_x^2 + p_z^2}, \quad \theta = \arccos(p_z/r) \text{ (co-latitude)}, \quad \lambda = \text{lon.}$$

Schmidt quasi-normalized associated Legendre functions P_n^m and derivatives $dP_n^m/d\theta$, as coded (`wmm.c:178`): seeds $P_0^0 = 1$, $P_1^0 = \cos \theta$, $P_1^1 = \sin \theta$; for $n \geq 2$,

$$P_n^n = \sqrt{\frac{2n-1}{2n}} \sin \theta P_{n-1}^{n-1}, \quad P_n^m = \frac{2n-1}{\sqrt{n^2-m^2}} \cos \theta P_{n-1}^m - \sqrt{\frac{(n-m-1)(n+m-1)}{(n-m)(n+m)}} P_{n-2}^m,$$

the second covering zonal ($m = 0$), sub-diagonal ($m = n - 1$, second term 0), and tesseral terms; derivatives follow by differentiating the same recursion.

Field synthesis (geocentric NED), $a = 6371.2$ km:

$$\begin{aligned} X_{gc} &= \sum_{n=1}^{12} \left(\frac{a}{r}\right)^{n+2} \sum_{m=0}^n (g_n^m \cos m\lambda + h_n^m \sin m\lambda) \frac{dP_n^m}{d\theta}, \\ Y_{gc} &= \frac{1}{\sin \theta} \sum_{n=1}^{12} \left(\frac{a}{r}\right)^{n+2} \sum_{m=0}^n m (g_n^m \sin m\lambda - h_n^m \cos m\lambda) P_n^m, \\ Z_{gc} &= - \sum_{n=1}^{12} (n+1) \left(\frac{a}{r}\right)^{n+2} \sum_{m=0}^n (g_n^m \cos m\lambda + h_n^m \sin m\lambda) P_n^m. \end{aligned}$$

Geocentric $\rightarrow$ geodetic NED rotation by $\delta = \varphi - \varphi_{gc}$ ($\varphi_{gc} = \pi/2 - \theta$):

$$X_{NED} = X_{gc} \cos \delta + Z_{gc} \sin \delta, \quad Y_{NED} = Y_{gc}, \quad Z_{NED} = Z_{gc} \cos \delta - X_{gc} \sin \delta.$$

Declination (`wmm_declination`, `wmm.c:251`): $D = \text{atan2}(Y_{NED}, X_{NED})$. The horizontal magnitude $H = \sqrt{X^2 + Y^2}$ and vertical Z supply the §4.9 magnetic invariants.

As-implemented. Secular variation is linear extrapolation from the epoch coefficients (valid over the model's 5-year validity window). `wmm_load` requires exactly the 90 degree-12 coefficient rows and rejects truncated files. Powers $(a/r)^{n+2}$ are formed by iterative multiplication, not `pow`. Poles are guarded by clamping $\sin \theta \geq 10^{-10}$.

Source: WMM2025 Technical Report, Chulliat et al. 2024 (NCEI/BGS) — the defining document; ALF recursion Langel 1987 / WMM Technical Note 28 (*code comment*); WGS-84 NIMA TR8350.2 (*canonical*).

14. Symbol $\leftrightarrow$ code map

Symbol	Meaning	Code
q	attitude quaternion, body $\rightarrow$ NED, scalar-first	<code>mekf_t.q[4]</code>
b	gyro bias	<code>mekf_t.bias[3]</code>
P	6 $\times$ 6 error covariance $[\delta\theta \mid \delta b]$	<code>mekf_t.P[6][6]</code>
$R(q)$	body $\rightarrow$ NED rotation matrix	<code>q_to_R()</code>
Q_g, Q_b	process-noise variances / step	<code>mekf_t.Qg, .Qb</code>
R_a, R_m	accel / mag meas. noise var	<code>mekf_t.Ra, .Rm</code>
H	measurement Jacobian	<code>eskf_update, eskf_update_yaw</code>
γ	χ^2 innovation gate	<code>ACCEL_CHI2_GATE=11.34, YAW_CHI2_GATE=6.63</code>
m_{ref}	Earth-field reference, NED, Gauss	<code>mekf_t.m_ref[3]</code>
q_{quiet}	quiescence EMA \$(	<code>a</code>
g_{body}	accel gravity direction, body	<code>mekf_t.g_body[3]</code>
H_s, T_z	sig. wave height, zero-cross period	<code>seastate_wave_height/_period</code>

Symbol	Meaning	Code
a_D	NED-down linear accel	<code>heave_update</code>
N, B	Allan noise density, bias instability	<code>allan_characterize</code>
g_n^m, h_n^m	Gauss coefficients	<code>wmm_t.g/.h, .g_sv/.h_sv</code>
D	magnetic declination	<code>wmm_declination</code>

15. References

Citations are to the canonical source for each method; imud implements standard techniques rather than novel mathematics. Items the maintainer may wish to verify against a specific edition/page are flagged [verify].

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*This document reflects the code at the release it ships with. When a numerical routine changes, update the corresponding section and its **As-implemented** note in the same commit.*